

'The line between intervention and abuse' – autism and applied behaviour analysis

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Abstract

This article outlines the emergence of ABA (Applied Behaviour Analysis) in the mid-20th century, and the current popularity of ABA in the anglophone world. I draw on the work of earlier historians to highlight the role of Ole Ivar Lovaas, the most influential practitioner of ABA. I argue that reception of his initial work was mainly positive, despite concerns regarding its efficacy and use of physical aversives. Lovaas' work, however, was only cautiously accepted by medical practitioners until he published results in 1987. Many accepted the results as validation of Lovaas' research, though both his methods and broader understanding of autism had shifted considerably since his early work in the 1960s. The article analyses the controversies surrounding ABA since the early 1990s, considering in particular criticisms made by autistic people in the 'neurodiversity movement'. As with earlier critics, some condemn the use of painful aversives, exemplified in the campaign against the use of shock therapy at the Judge Rotenberg Center. Unlike earlier, non-autistic critics, however, many in this movement reject the ideological goals of ABA, considering autism a harmless neurological difference rather than a pathology. They argue that eliminating benign autistic behaviour through ABA is impermissible, owing to the individual psychological harm and the wider societal impact. Finally, I compare the claims made by the neurodiversity movement with those made by similar 20th- and 21st-century social movements.

Keywords

Applied Behaviour Analysis, autism, behaviourism, Ole Ivar Lovaas, neurodiversity

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Proponents of Applied Behaviour Analysis (ABA) have praised it as the only scientifically validated method of treating or curing autism, while its critics have vehemently condemned it as abusive.¹ This article analyses these debates, outlining the rise of ABA from an experimental therapy to its contested position today as the ‘gold standard’ of autism treatment. I argue that early critics of ABA raised concerns only with regard to its effectiveness or use of aversives, while more recent ones have additionally focused on the curative aims of ABA. These latter critics are often autistic self-advocates who are part of the ‘neurodiversity movement’; they reject the medical model of autism and, therefore, the need for any treatment whatsoever. Paying particular attention to these arguments, this article uses accounts written by autistic people as its primary sources, alongside others such as scientific articles, popular press articles and interviews. While many academics have examined autistic autobiographies, I instead use informal texts such as blog posts written by autistic people. These are used to avoid a common over-reliance on the accounts of those autistic people with sufficient resources to publish autobiographies, but the use also reflects the tendency of autistic people to discuss ABA online rather than through formally published works. These debates are situated within the histories of autism and behaviour therapy: historians of both fields have seldom focused on ABA, despite its current prominence in both behavioural and autism interventions. This article also compares the neurodiversity movement’s criticism of ABA to other 20th- and 21st-century social movements that similarly reject pathologization, drawing new parallels with the gay rights movement and mental health service user organizations. Existing accounts of ABA are either brief asides in broader accounts of autism (for example, Eyal, 2010; Murray, 2012; Waltz, 2013) or non-academic accounts that are rich in factual detail but lack historical context (Donvan and Zucker, 2016; Feinstein, 2011; Silberman, 2015). This article is therefore the first in the historical literature to take ABA as its focus.

The language used in discussions of ABA and autism is contentious. Many autistic people reject the use of person-first language, arguing that it suggests that they are divisible from their autism, which they consider an inseparable part of their personhood; I therefore opt for ‘autistic person’ rather than ‘person with autism’. Similarly, the appropriateness of medical terms such as ‘treatment’ and ‘therapy’ is disputed by those arguing that autism is not a disease. Where there are no suitable alternatives these words are used in this article, but the – often unchallenged – assumptions such terms rely on should be noted. Relatedly, definitions of autism and ABA have changed, as will be discussed in detail. The diagnostic category has expanded and shifted, meaning many people with a diagnosis today have ‘symptoms’ milder than or different from those in the 1960s. This has led some to label autism a ‘moving target’ (Hacking, 2013). Others have concluded that the concept has no meaningful relevance (Timimi, Gardner and McCabe, 2011) as there remains no consensus on autistic diagnostic criteria, and even less on the existence or type of an underlying aetiology.² Autism itself is therefore as highly contested a condition as ABA is a therapy. This article does not attempt to evaluate the ontological status of ABA or autism (see Hacking, 1999), but aims instead to shed some light on the shifting debates surrounding these categories.

What is ABA today?

The status of ABA today is complex and varies significantly by geographical region. However, in North America at least, ABA is currently very popular. There are no statistics revealing the exact number of people to have undergone the therapy, but it is clearly widely respected by much of the American medical industry. Influential autism-focused organizations, including the world's largest autism charity, Autism Speaks, are particularly vocal supporters. Autism Speaks (n.d.) recommends ABA above all other alternatives, describing how its 'scientifically validated approach' leads to 'significant improvements'. Other introductory guides to autism call ABA 'the only proven treatment' (Autism Truths, n.d.) and 'the best method for managing autistic behaviour' (Autism, n.d.). Anybody researching autism following a recent diagnosis will quickly find recommendations for ABA.

The persuasiveness and ubiquity of these recommendations have led to increased demand for ABA, as evidenced by the number of parents lobbying for it to be available. Many have brought legal action against insurance providers who do not cover the cost of ABA (Freudenheim, 2004). As a result, 38 states introduced laws between 2010 and 2014 requiring insurance companies to cover ABA-related costs (Allure, 2014); legal battles continue in states without such laws. In the UK, meanwhile, schools utilizing ABA have been at maximum capacity, ostensibly owing to the popularity of the therapy (Lambert, 2013). Parents have also campaigned for local councils to provide ABA on demand (Benyon, 2014). Those parents who have accessed ABA for their children have an 'overwhelmingly positive' opinion of ABA (McPhilemy and Dillenburger, 2013: 154), which likely leads to many parent-to-parent recommendations for ABA. Autism charities also fund ABA research to be conducted by well-established researchers at prestigious institutions. These charities and researchers enlist celebrity supporters to further raise awareness of ABA (USCB, n.d.). On both sides of the Atlantic, universities teach ABA through dedicated degree courses. This ever-growing institutional support for ABA is fuelled by, and in turn fuels further, the efforts of campaigning parents and charities.

This dominance of ABA is particularly unusual because purely behavioural treatments are not generally in medical favour today. Many behaviours once subject to behavioural intervention are no longer considered illnesses in western medicine, such as homosexuality. Among conditions that are still medicalized, such as depression and anxiety, intervention usually involves a blend of cognitive and behavioural models through cognitive behavioural therapy (CBT) (Marks, 2015). This contrasts with ABA, which remains rooted in traditional behaviourism.

What, though, is this popular treatment? Proponents of ABA broadly agree that it is characterized by the breaking down of behaviour into three components (Autism, n.d.). The first of these is the antecedent, which is whatever triggered or preceded the behaviour being analysed. The second is the behaviour itself, and last is the consequence of the behaviour, which may be positive or negative. The practitioner controls this using rewards and punishment. Through this, ABA practitioners aim to extinguish 'undesirable' behaviours and to promote 'pro-social' ones, such as increased eye contact or verbal communication. This logic, which has appealed to many academics, parents and charities, originated in the mid-20th century.

‘Deliberate, calculated harshness’: The origins of ABA

ABA emerged from behaviourism, a school of thought that developed in psychology from the 1910s (Smith, 2013: 140–1). Early behaviourist researchers were primarily concerned with conditioning in animals, drawing on the work of earlier researchers such as James Watson and Ivan Pavlov. The most famous of these was B. F. Skinner, who became well known for rewarding rats that exhibited a certain behaviour with food; this method became known as ‘operant conditioning’ (Skinner, 1938). The underlying assumption – that behaviour could be modified through rewards and punishment – was extended to humans by a small number of researchers from the 1950s, particularly in Britain. Practitioners began to take a behaviourist approach to the treatment of mental disorders at the University of London’s Institute of Psychiatry and the Maudsley Hospital (Marks, 2015). Psychiatrists such as Isaac Marks and Michael Gelder, and clinical psychologists such as Monte Shapiro, Stanley Rachman and Hans Eysenck (Eysenck and Rachman, 1965) deployed techniques such as aversion, exposure or operant conditioning to treat neurotic disorders such as phobia, obsessions and compulsions (Buchanan, 2010).³

By contrast, a number of behaviourist researchers in America focused particularly on lifelong rather than emergent conditions. University of Indiana researcher Paul Fuller, for example, attempted to use positive reinforcement on adults with severe learning difficulties or, as his paper’s title called them, ‘vegetative human organisms’ (1949: 587). Others, such as Ayllon and Michael (1959), studied patients with seemingly irreversible cases of schizophrenia. Such work consisted largely of exploratory investigations, and most of these researchers spent the majority of their careers pursuing other interests, undertaking behaviourist studies only occasionally.

Behaviourist research into learning difficulties was also limited by a widely held assumption that people with learning difficulties were beyond help.⁴ At the time, individuals were frequently institutionalized for life and would rarely be assessed by clinicians after their admittance. Moreover, there were proportionately fewer people diagnosed with learning difficulties; the label of autism, for example, was reserved for those with the most ‘severe symptoms’. Researching treatments for these patients was therefore considered both futile and largely irrelevant to psychological and psychiatric practice. Even those researchers who did study the area accepted this. For example, Indiana researcher Charles Ferster, a close collaborator of B. F. Skinner, began his 1961 paper by admitting its limited epidemiological importance (Ferster, 1961: 437). This belief, combined with the small number of researchers in this field, meant that there was no unified approach to such work; behaviourist practices varied considerably and practitioners disagreed about the causes of the behaviour they were treating (Smith, 1997: 650).

Ferster’s paper was the first to investigate autism through a behaviourist lens. In the experiment Ferster locked an autistic child in a room daily for a year (Ferster, 1961). He refused to attend to the child when it cried, and its tantrums stopped. He concluded that parents had exacerbated their children’s autism by responding to their tantrums. Such medical parent-blaming was common; over-attentive mothers had been denounced as ‘smother-mothers’ while those who failed to give their child enough attention were

'refrigerator mothers'.⁵ The children of these women, it was argued, responded by developing autism. Ferster, though, was not primarily interested in autistic children, but hoped that his research would shed light upon schizophrenia, of which autism was considered a childhood form. In keeping with a medical profession largely uninterested in childhood learning difficulties, then, the first behaviourist investigation of autism was actually concerned with an adult condition.

As well as the University of Indiana, where Ferster was based, the UCLA psychology department was one of the few places where researchers used conditioning to study intellectual disabilities. It was here that Sidney Bijou coined the term 'behaviour analysis' and divided behaviour into antecedents and consequences; this division would later become the hallmark of ABA. Under Bijou, the consequences were verbal praise and criticism, though researchers working elsewhere with animals had begun to use electric shocks. UCLA was also where Ole Ivar Lovaas became assistant professor in 1961.⁶ Lovaas intended to study psychoanalysis but was soon captivated by the work of Bijou and his colleagues, and instead opted to study behaviourism.

Unlike Bijou, Lovaas considered verbal consequences alone insufficient, particularly when dealing with non-verbal children. Lovaas decided the child must first acquire communication skills before being able to respond to verbal feedback. Lovaas also wanted to work with autistic children as he considered them a blank slate: '[Y]ou start pretty much from scratch when you work with an autistic child . . . they are not people in the psychological sense' (Lovaas, 1974: 76). This belief was not unusual, mirroring Fuller's description of 'human vegetative organisms'. Lovaas initially worked with an autistic child called Beth (ibid.). At first, he used only rewards but became concerned when Beth reverted to self-injurious behaviour. Additionally, he noted that Beth struck her head against the sharpest available surface. He concluded that Beth was deliberately sabotaging his experiment and beat her until she stopped. Seeing that this stopped her self-injury, Lovaas began to use such 'aversives' regularly, initially through beatings and later through shocks and the withholding of food (Lovaas, 1966).

There was initial skepticism towards Lovaas' approach. Skinner, for example, had earlier argued that the use of punishment was associated with undesirable side effects (Dixon, Vogel and Tarbox, 2012: 7). The curative aims of the work were unquestioned. Lovaas pre-empted criticisms of the method's harshness to other researchers, arguing that the use of physical aversives was better than the alternative of institutionalization. Moreover, Lovaas was not the only researcher advocating the use of physical aversives, merely the best known; others had concurrently been trialling the use of shock-therapy for autistic children (Risley, 1968). Persuaded by the arguments made by Lovaas and others, even initially skeptical academics, such as parent-researcher Bernard Rimland, eventually supported their methods in at least some cases.

Lovaas additionally courted broader media attention, inviting journalists to watch his treatment. In press interviews he admitted his methods were harsh, but defended them as the only way to improve the child's life. Such defences proved persuasive to journalists, as demonstrated by a 1965 multi-page report in *Life* magazine. Featuring photographs of children being hit, the report acknowledged the treatment was 'shocking', 'drastic' and 'deliberated, calculated harshness', but was nonetheless supportive, praising sensible 'old-fashioned' ideas, the 'loving attention' of the researchers and the way it connected

with 'far-gone mental cripples'. The article depicted Lovaas' electric shock room, reassuring readers that it was a last resort, used for children who did not respond to reprimands owing to a lack of anxiety: 'To give [a child] something to be anxious about, [they are] taken to the shock room' ('Author Unknown', 1965: 90). This teaches them a 'new respect' for the word 'no'. Aware this might still seem unduly harsh, the article emphasized the 'patience and tenderness' of the researchers. Lovaas, through such media engagement, was therefore successfully promoting his work.

By the late 1960s many American researchers were working in ABA and in 1968 the first issue of *The Journal of Applied Behaviour Analysis* was released. Researchers also formed professional societies such as the Association for the Advancement of Behavioural Therapy and the Behaviour Therapy and Research Society (Horwitz, 2013). In general, this work focused on autism, but some researchers deployed ABA towards other conditions. A particularly famous example was the 'Feminine Boy Project', an attempt to preemptively 'cure' those male children considered at risk of becoming gay or transvestites.⁷ The leaders of the project sought advice from Lovaas, and used an approach strongly influenced by ABA, indicating Lovaas' status in the field. Though others had developed the ideas behind ABA, and there were numerous active ABA researchers, Lovaas was clearly one of the most well respected. His eye-catching work and media engagement had ensured him a pre-eminent position among practitioners.

However, Lovaas was not uniquely popular outside of behaviourism, in wider autism and medical circles. Though ABA was becoming increasingly popular, psychoanalytic accounts of autism were still more widespread at this point, owing to the work of researchers such as Bruno Bettelheim and his colleagues who had made similar claims of 'curing' autism (Dolnick, 1998: 212). Parents of autistic children, though, seemed particularly interested in Lovaas' work, even though it was not yet in the medical mainstream. Bernard Rimland, promoting Lovaas' work in 1966, found many parents were already aware of ABA (Silverman, 2012). Lovaas was invited to the first meeting of the National Society for Autistic Children (NSAC) in 1969; the organization's first president, Ruth Sullivan, personally supported his work (quoted in Feinstein, 2011: 130).

Two main factors explain this enthusiastic parental support. First, few people were researching autism at this time, and many still considered autistic children beyond help. Although psychoanalytic therapies were increasingly available, parents still had sufficiently few options that they welcomed Lovaas' work even when many medical practitioners did not. Second, non-behaviourist medical practitioners supported parent-blaming theories such as the 'refrigerator mother' hypothesis. They advocated treatments that often involved taking children away from their parents, which was dubbed a 'parentectomy' by some therapists (Dolnick, 1998). Lovaas' approach, contrastingly, actively involved parents as what historian Chloe Silverman (2012: 113) has termed 'explicitly trained cotherapists'. This involvement of parents rather than their deliberate exclusion made Lovaas' ideas more appealing to parents, particularly as its methods were easy to apply at home (Nadesan, 2005). Nonetheless, in more general terms, behaviourist understandings of autism were only gradually eroding the hegemonic hold of the psychoanalytic framework.

Throughout the 1970s and 1980s Lovaas and his colleagues continued their research as the field of ABA grew. During this time Lovaas worked with younger children,

stressing the importance of intensive, early intervention. This brought additional criticism; some researchers were concerned that 40 hours per week of treatment was excessive. Lovaas also dropped the use of many physical aversives, such as electric shock treatment, during this period, as support for their use waned during this time. NSAC had publicly supported the use of physical aversives as late as 1975 but by 1988 effectively called for a ban on their use (Silverman, 2012). In the same year Anne Donnellan published a book, *Progress without Punishment* (Donnellan, 1988), that condemned the use of physical aversives, further demonstrating increasingly widespread discomfort. Some, such as Bernard Rimland, continued to support physical aversives, particularly in cases of self-injurious behaviour. Nonetheless, Lovaas' discontinuation of the harshest aversives made his methodology more acceptable to the wider medical establishment, particularly as concerns about child welfare had become stronger than they had been in the 1960s.

Lovaas' work also became more acceptable to the medical profession as behaviourism's competing theory, that of psychoanalysis, became less influential. Social historian Majia Holmer Nadesan has suggested that this was due to the difficulties in quantifying the effects of psychoanalysis, while historian Edward Dolnick argued that the psychoanalytic, parent-blaming school of thought 'wasn't demolished in a single blow' (Dolnick, 1998: 225) but lost influence gradually owing to a number of factors. One of those identified by Dolnick is the work of American researchers such as Marian DeMyer who conducted intelligence tests on autistic children (DeMyer, Barton, Alpern *et al.*, 1974). They concluded that autistic children were fundamentally 'retarded', implying a biological basis for autism. This undermined dominant psychoanalytic thought at the time; autistic people were not believed to differ neurologically, but were thought to have normal or even above average intelligence that was hidden by an autism caused by factors such as parenting. As Dolnick argues, this made the position of Bettelheim and his colleagues increasingly untenable (Dolnick, 1998: 227). Behaviourism was a readily available alternative framework.

As well as the wider shift in the psy-disciplines away from psychoanalysis, academic understanding of autism specifically changed dramatically in the 1970s and 1980s. It was increasingly understood as something present at birth, and was characterized in different ways, as historian Bonnie Evans (2013) has argued. During the 1960s autism was associated with psychosis, fantasy and schizophrenia, but by the 1970s researchers such as Michael Rutter were arguing that 'the autistic child has a deficiency of fantasy rather than an excess' (Rutter, 1972: 327, quoted in Evans, 2013: 4). Using epidemiological methods, Rutter and his colleagues placed less emphasis on interpreting behaviour as psychotic, and looked instead at language and social impairments. The long-standing link between autism and schizophrenia therefore disintegrated, as shown when childhood schizophrenia was removed from the third edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-III) and replaced by categories of autism.

By the late 1980s, therefore, ABA and autism itself had been radically transformed. It was in this new environment that Lovaas published findings in 1987 claiming that 47% of his patients had recovered from autism. In this study, Lovaas compared the 'intellectual and educational functioning' of two groups of children under the age of four that had been diagnosed with autism by multiple professionals. The first group of 19 children underwent 40 hours of one-to-one behavioural intervention weekly for two years; 47% of these were found to have 'normal functioning' after this time, compared with only 2%

of the 40 children in the control group who had received only 10 or fewer hours a week of intervention over the same time period (Lovaas, 1987). These developments were soon widely known and were taken as proof that ABA was effective; such evidence had not previously been published. Both proponents and critics of ABA acknowledge the importance of this result in establishing the widespread popularity of ABA.

The benefits of ABA were additionally detailed in *Let Me Hear Your Voice* (Maurice, 1994), an emotional account of a family 'triumphing' over autism for which Lovaas wrote the foreword; this brought the therapy greater fame. ABA became strongly associated with those stories, rather than with its earlier use of physical aversives. It was also part of a new medical view that encouraged parents to fix childhood autism, and no longer blamed them for causing it. Crucially, this occurred just before autistic diagnosis rates increased dramatically throughout the 1990s and early 2000s (Wing and Potter, 2002). With more and more children being diagnosed as autistic, there were ever-growing numbers of parents searching for treatments. ABA, marketed to parents as a small number of simple steps, bore similarities to widespread parenting and teaching methods of reward and punishment, allowing it to be seen as an apparently intuitive approach. This, along with ABA's involvement of parents, its reduced harshness, its claims of scientific validation and its stories of triumph, ensured that it was soon widely regarded as the best treatment for autism.

Continuing controversy: ABA in recent years

As ABA became more popular it attracted more controversy. Part of this has been academic; scientists have continued to argue that the benefits of ABA have been overstated (Jordon, 2001; Olsson *et al.*, 2015; Ospina *et al.*, 2008; Weiss, 2006). Several have attempted, and failed, to reproduce Lovaas' results (Mudford *et al.*, 2001; see also Matson and Neal, 2012: 12) while others have highlighted the seemingly low quality of ABA research; The Institute of Education Sciences concluded there was little high-quality evidence for the Lovaas model. Still others have criticized the lack of randomized controlled trials (RCTs) in ABA research (Howlin, 2005), which are considered by many to be the gold standard of evidence-based medicine.

ABA has always, however, faced some academic disapproval, though this has varied in degree and kind from the 1960s.⁸ Since the early 1990s, though, ABA has faced a new challenge from autistic people themselves. Rather than focusing on the effectiveness of the therapy, these opponents have argued against the need for any therapy at all; they consider ABA part of a system that mistakenly considers autism a disease and therefore take issue with the aims of ABA, though they have also led campaigns against the use of physical aversives. Such critiques have emerged as autistic people have increasingly been involved in debates about autism, its categorization and its relation to medicine. These activists and their allies constitute the 'autism rights' or 'neurodiversity' movement that has developed in the past 25 years.

Before this, organizations focused on autism did exist, notably the Autism Society of America (ASA) and the National Autistic Society (NAS) in the UK, both of which were founded in the 1960s.⁹ Importantly, these were mainly run by parents of autistic people;

autistic people themselves were therefore excluded from debates regarding autism (Dawson, 2003). From the late 1980s this changed, first through the work of Temple Grandin, an autistic woman who spoke about autism at conferences and in books (Grandin, 1986). Soon after, a group of American self-advocates established autistic-run organizations, both online and off-line, as well as retreats and conferences (Sinclair, 2005). In the UK, too, autistic people gained new platforms, working with the NAS to launch a publication written and edited by autistic people (*Aspergers United*). Individually, autistic people wrote blogs, autobiographies and speeches that reached wide audiences, while collectively they established advocacy groups such as the American Self Advocacy Network (ASAN, n.d. (a)).

Though a diverse group, members of the neurodiversity movement share certain beliefs. They argue that problems commonly understood as being caused by autistic behaviour actually stem from a lack of tolerance among non-autistic people. Rejecting the medical model of autism, they consider it a neutral difference rather than a deficit (Kapp, 2013). In doing so they draw parallels between the spectrum of neurological functioning and that of sexuality, arguing that the medicalization of either is wrong. This argument has been made as the psy-disciplines have increasingly employed cognitive models, which emphasize different types of brain ‘wiring’ or processes, rather than discrete pathologies (see Boden, 2006; Gardner, 1987; and Smith, 2013: 243–8). As Nadesan has argued (2005: 112), this has ‘problematised old boundaries between normalcy and abnormality’ and replaced ‘categories of normalcy and pathology with a multitude of developmental continua . . . [leading] to a more nuanced’ understanding of ability and intellect.¹⁰ This has lent some scientific credence to the neurodiversity movement; neurodiversity supporters’ understanding of autism as a neutral, nuanced difference seems more readily compatible to cognitive metaphors of complex, alternate wiring systems or differences in ‘processing’, than to earlier psychoanalytic ideas of trauma, or to the behaviourist emphasis on positive and negative actions. Members of the movement have campaigned on a range of issues, including the relationship between medicine and autism. Unsurprisingly they have often targeted the best-known medical treatment for autism: ABA.

Though they reject the curative aims of ABA, some self-advocates have nonetheless also highlighted its physically abusive qualities. The Autism Friends Network of autistic self-advocates, for example, states that it is opposed to ‘physically . . . harmful “treatments” targeting autistic people’, suggesting they would be called ‘torture’ if applied to non-autistic children (Autism Friends Network, 2013). Autistic activist Julia Bascom (2012: 178) has similarly said that ‘when you’re autistic [ABA] is not therapy, it’s abuse’. A former ABA practitioner has argued a similar point, saying that the continued use of physical aversives is abusive. Echoing autistic people who have undergone ABA, she concludes that it can leave people ‘traumatised and devalued’ (Socially Anxious Advocate, 2015).

Importantly, these are criticisms of recent and continuing practices. This is well illustrated by the campaign led by ASAN against the Judge Rotenberg Center (JRC).¹¹ This centre has repeatedly been embroiled in controversy since the early 1990s, having been investigated by its own governors, the American Food and Drug Administration and the UN Special Rapporteur on Torture. Autistic activists argue that there have been a ‘slew of preventable deaths’ in the facility (Neumeier, 2012: 209).¹² For example,

autistic self-advocate Shain Neumeier (*ibid.*) alleges that a young woman died of a perforated stomach ulcer in 1990 after staff allegedly mistook her reactions to the pain as misbehaviour and consequently subjected her to physical aversives. This allegation has often been repeated online by self-advocates, who see the case as a particularly powerful example of the dangers of ABA. More recently the JRC has become well known as the last remaining American institute where electric shock therapy is used. ASAN and its supporters have vigorously campaigned against JRC for this reason and have criticized organizations that support it, including the Association for Behaviour Analysis.

Critics also argue that ABA damages patients psychologically, even when physical aversives are not used. They suggest ABA practitioners discourage behaviour that is itself a form of communication; for example, 'if an autistic child who screams every time he is taken to the supermarket is trained not to . . . he may still be experiencing pain from the fluorescent lights and crush of strangers' (Harmon, 2004). Other self-advocates have supported this view, saying that ABA 'takes away our voice' (Bascom, 2012: 182). Moreover, activists suggest that ABA practitioners wrongly focus on inhibiting unusual but self-soothing behaviour. Many autistic people, for example, engage in 'stimming', which refers to movements such as flapping hands that they find therapeutic, but which ABA practitioners discourage. Self-advocates have therefore denounced ABA as abusive on psychological grounds, as well as physical ones. Indeed, it is these self-advocates that have in particular deployed the term 'abuse', tapping into widespread public concern with child abuse that did not exist in the 1960s. This characterization of ABA parallels a more general shift and expansion of the concept and categories of child abuse (see Hacking, 1991).

Some activists believe the issue of physical abuse has distracted society from these deeper issues within ABA. Michelle Dawson, an influential Canadian autistic activist, has argued that ABA is not subject to the ethical scrutiny it warrants because of faulty assumptions that ethical concerns have been 'dealt with' now that physical aversives are rarely used. Dawson publicly argued this, submitting testimony to a Canadian court arguing against the provision of ABA; the court ruled in agreement with Dawson, citing her testimony in their decision.¹³ Her essay 'The Misbehaviour of Behaviourists – Ethical Challenges to the Autism – ABA Industry' has been particularly influential, and is widely cited by those in the neurodiversity movement (Dawson, n.d.). In it, Dawson criticizes the arguments ABA proponents have used to bypass ethical considerations; for example, by arguing that taking consent is unnecessary owing to the severity of autism. Comparisons between autism and cancer, and between ABA and chemotherapy, are common in such justifications, and represent the medicalized view of autism Dawson and other critics reject. Dawson believes it is this view that has led to a lack of RCTs in ABA research; it is considered immoral to withhold ABA, the best-known treatment, from children in a control group, meaning full RCTs cannot be conducted.¹⁴ Such decisions, she concludes, cannot be justified without first acknowledging that ABA needs fuller ethical investigation.

Dawson and other self-advocates take especial issue with the aims of ABA to inhibit or reduce 'autistic behaviour' (e.g. see Dawson, n.d.). Many autistic people see such behaviour as authentic and harmless representations of their personality, arguing that it is

only a lack of societal understanding that renders it problematic. ASAN (n.d. (b)), for example, has stated that ‘autistic children should not have to grow up constantly being told that their natural behaviours are wrong and they cannot be accepted as who they are’. Prominent self-advocate Amy Sequenzia has similarly criticized ABA for unjustifiably holding up neurotypical behaviour as an ideal, arguing that its attempt to ‘change how an autistic acts, reacts or interacts with the world’ is wrong (Sequenzia, 2015). As an anti-cure ally of the movement has summarized, ‘ABA can only work if we view autism as a deficit and something that must be removed. Autism is a complex neurological difference that cannot be moved . . . I don’t agree with the basic tenets of ABA because it is a methodology based in looking at those it treats as deficient and inferior’ (Zurcher, 2012). Some researchers share those concerns; the head of autism research at the London Institute of Education worries about ‘the underlying ideology of ABA – to make them indistinguishable from their peers’ (Lambert, 2013). This, she believes is actively harmful. Similarly, Dawson has characterized a societal support of ABA to ‘being in a society where “blacks have rights so long as they become white”’. Amanda Vivian (2012) has concluded that ‘it’s wrong to say “cure autism now”’. Of course it’s wrong to say autism is a tragedy, a disease.’ The implication of these arguments is that autistic disability is better understood through a social rather than medical model.

Given this, activists also denounce other therapies. Treatments condemned by activists and dismissed as ineffective by scientists include transcranial magnetic stimulation (Oberman, Rotenberg and Pascual-Leone, 2015), dolphin therapy (Marino and Lilienfeld, 2007), and massage therapy (Lee *et al.*, 2011), as well as the use of drugs, though this is still common (Broadstock, Doughty and Eggleston, 2007). In North America critics have vociferously campaigned against chelation, the removal of heavy metals from the bloodstream, following a number of deaths (Willingham, 2012). In France, similar campaigns have been launched against ‘packing’, the wrapping of autistic children in cold, wet sheets (Spinney, 2007). Nonetheless, ABA is most frequently targeted by self-advocates owing to its popularity and its perceived scientific support.

Some ABA supporters have responded to the allegations levied by the neurodiversity movement. Christina Burkaba (n.d.), for example, believes the critics are confusing the ‘solid science’ of ABA with Lovaas’ work, which she admits is flawed. She denies that the normalizing aims of ABA are problematic, arguing that changing behaviour is ‘the whole point’, as in all education. Richard Hastings (2013), a psychology professor at Warwick University, similarly says it is ‘people, not approaches or theoretical positions, that abuse other people’, implying that ABA itself should be defended. Such direct engagements with the arguments of the neurodiversity movement are rare, however; most proponents of ABA are enthusiastic and uncritical supporters of the therapy, as previously indicated. This lack of meaningful dialogue between supporters and detractors of ABA means that these controversies are unlikely to end soon.

These debates are, of course, shaped by their national setting. Notably, ABA is much more popular in the USA, as outlined at the beginning of this article. In the UK only a few organizations, small in number and size, work to promote the therapy and the largest organization, the NAS, refuses to endorse it. Its website states only that there is contradictory evidence regarding ABA (NAS, n.d.) and suggests critical ways of approaching the claims made about its effectiveness. It also reminds readers that many autistic people

do not consider autism a disease. This contrasts with the American Society of Autism, which, though less supportive of ABA than newer organizations such as Autism Speaks, does not question the curative aims of the therapy. The historical factors explaining these differences are numerous and deeply inter-connected.

First, ABA was developed in the USA, meaning it was able to spread more quickly through institutions in that country. Lovaas had enthusiastically promoted the therapy in academic and media spheres, aided by supporters such as Catherine Maurice and Bernard Rimland. These proponents did not have equivalents in the UK, meaning ABA gained less support, particularly when the evidence alone was not considered conclusive. Broader differences in healthcare culture between the USA and the UK have also had an impact. The USA has a relatively open healthcare market, contrasted with the UK's nationalized health service. Individual practitioners are therefore more freely able to promote their therapies through advertising and media engagement. In the UK, the NHS is obliged to supply only 'cost-effective' treatment, which is likely to preclude ABA, a resource-intensive therapy. America's private model of healthcare funding is therefore more likely to fund ABA, particularly as more is spent on healthcare per person in total. The strong support for ABA in the USA is therefore likely to be part of a larger systematic difference that allows a wider range of therapies to be given consideration in medical spheres.

Other institutional forces have shaped this debate. The USA has a large number of autism-focused organizations competing for funding and media attention. Newer ones, notably Autism Speaks, have eclipsed less aggressively pro-cure organizations such as the ASA in both these areas. In the UK meanwhile, the NAS is unrivalled as the national expert organization for autism. Its reluctance to endorse ABA has therefore limited ABA's popularity, and is typical of the NAS's more neurodiversity-friendly approach. The NAS has, in turn, taken this position because of several historical connections, in particular its close links to the Institute of Psychiatry and the Maudsley Hospital (Evans, 2014). Researchers at those institutes in the 1960s were less focused on curing autism, as Bettelheim and Lovaas were aiming to do at the same time, but instead emphasized the educational policy relevance of their work (*ibid.*). Findings from those institutes were used to lobby the government to provide education for autistic children, a move that emphasized their potential to learn rather than their need to be cured. Again, these differences link to broader cultural differences in the USA and the UK, but provide some explanation as to why the ABA debates have manifested in such different ways across the Atlantic.

Comparisons with other social movements

The criticisms of psychiatric practices and thinking made by autistic self-advocates have parallels in other social movements based around healthcare. Most similar are those which also reject an entirely medical, pathologized understanding of their condition. Surprisingly, few of these groups are referenced by autistic self-advocates in their publications, despite common ways of understanding, similar chronologies and shared opponents.

The activism of the parents of autistic people, particularly the parent-led activism of the 1960s, attempted to influence, rather than reject, the medical model. Since the 1960s,

parents have positioned themselves as autism experts, and were influential in discrediting 'parent-blaming' models of autism. However, self-advocates often position themselves in direct opposition to such parent-led organizations (Dawson, 2003). Though parents of autistic people did challenge medical expertise, they retained a medicalized model of autism, and de-stigmatized themselves while contributing to an othering of autistic people (Waltz, 2013). This is in sharp contrast to the self-advocate movement, which has eschewed the pathologization of autism and has often been in direct conflict with parent-led groups that continue to emphasize the expertise of parents rather than autistic people.

Autistic self-advocates have suggested that they are more comfortably aligned with the gay rights movement; many having drawn parallels between the spectrums of neurological functioning and of sexuality. Arguments made by autistic self-advocates are also similar to those made by gay activists who campaigned for homosexuality to be de-pathologized in the 1970s, such as the assertion that homosexuality was medicalized only due to a societal discomfort with non-conformity (Faderman, 2015), mirroring more recent arguments made by autistic activists. Homosexuality and autism have older parallels, too; both were treated in the 1960s psychoanalytically. 'Close binding' mothers purportedly caused homosexuality in sons at approximately the same time as 'refrigerator mothers' were believed to cause autism. Activists from both groups have also specifically targeted behavioural intervention. In 1972, for example, gay activist Ronald Gold disrupted the Association for the Advancement of Behaviour Therapy to condemn the individually and societally harmful nature of the therapy (*ibid.*). These arguments are strikingly similar to those made by members of the neurodiversity movement more recently. Though the neurodiversity movement has links to other movements, too, it is likely that the members deliberately compare themselves to the LGBT+ movement owing to its relatively recent success in de-pathologizing behaviour. This makes it a more politically useful comparator than newer movements that may not be perceived as having reached the same levels of success, such as the Hearing Voices Network (Hearing Voices Network, 2016).

The Hearing Voices movement campaigns against the stigmatization of hearing voices, a phenomenon medically known as 'auditory hallucination' that is commonly associated with schizophrenia diagnoses. This movement, like the neurodiversity, mental health service users' and disability rights organizations, also grew in influence in the late 1980s; 1987 is considered a turning point (Woods, 2013) as the first hearing voices congress was held that year. The Hearing Voice Network starts from the 'premise that hearing voices is a normal variant of behaviour, much like left-handedness' (Blackman, 2001: 189). This clearly aligns with arguments made about the spectrum of neurological functioning; both arguments are likely to have been aided by an increasingly cognitive understanding of mental health. An interesting difference between the two groups, however, is that autistic self-advocates reject people-first language, referring to themselves as autistic people rather than people with autism; the converse is true for many people with a diagnosis of schizophrenia (Saks, 2012). This reveals different methods of de-stigmatization. 'People with schizophrenia' prefer that choice of phrase to, in their view, historical misconceptions that they are defined by their condition, implicit in the use of the term 'schizophrenic' (Living with Schizophrenia, 2017). By contrast, many

autistic people eschew people-first language because they have historically been seen as 'trapped' by their condition, arguing instead that it is a fundamental and inseparable part of them.

The disability rights movement similarly rejects an exclusively medical model of understanding. Notably, it is considered to have blossomed in the late 1980s (Shapiro, 1993), just a few years before neurodiversity groups emerged in the late 1990s. Both make similar arguments; that their members can succeed not despite, but because of, their differences. They also both emphasize the need for inclusion in debates about their condition. The phrase 'nothing about us without us', for example, became well known in the late 1990s owing to disability rights campaigners (Charlton, 2000; Werner, 1997) but is now used by many movements, including the neurodiversity movement; the self-advocate-run ASAN uses it as a motto.

Broader mental health service user groups also challenge medical authority, and privilege lived experience over professional expertise in a similar way. The late 1980s and early 1990s also saw a 'real flowering of service user action' in mental health groups (Campbell, 2005: 74). This is a diverse mix of organizations and viewpoints. Some work closely with medical establishments, while others, such as the anarchist Mad Pride movement, are more radical, and reject such involvement. Nonetheless, all tend to agree that, at least in part, societal changes in understanding are needed rather than simply changing unusual behaviour (Dellar, Curtis and Leslie, 2000). Some also reject the need for any treatment at all (*ibid.*), as a number of neurodiversity advocates have also done. Despite differences in mental health service user groups, they therefore mirror the neurodiversity movement insofar as they challenge medical authority and expertise. These examples, however, caution against historically homogenizing social movements based around de-pathologization, despite many similarities in their methodologies and beliefs.

Conclusion

ABA has changed significantly since its inception. Lovaas, drawing on the work of earlier behaviourists, developed an unashamedly harsh therapy in the 1960s. It was not until ABA had largely dropped such aversives, though, that it came to be seen as the treatment for autism. This success was aided by apparent experimental validation, seemingly intuitive methodologies, positive media representations and broader shifts in the understanding of autism and psychiatry.

The perceived faults of ABA have also changed; early criticisms focused on efficacy or the use of physical aversives. These challenges have continued, but recent, novel condemnations of ABA also dispute the curative aims of the therapy. Autistic self-advocates with this view have been able to make these arguments owing to the establishment of neurodiversity organizations and modern methods of communication in the form of online fora and campaign websites, in addition to the support of some large institutions such as the NAS. The emergence within the psy-disciplines, and wider public discourse, of cognitive models of understanding difference has also offered a legitimating scientific language, which has been appropriated by the movement as an alternative to both behaviourist reductionism and medicalized pathologization. By rejecting medical

models in this way, autistic activists share many similarities with other movements, including the earlier campaign to de-pathologize homosexuality and more recent groups such as the Hearing Voice Network. Like those groups, the neurodiversity movement has distanced itself from medicine, challenged pathologization and questioned traditional expertise, while at the same time having mobilized alternative expert discourses from cognitive psychology to support its agenda. In doing so, it has rejected ABA as unethical and, ultimately, unnecessary. It is unclear how influential this view has been beyond groups of the neurodiversity movement. In the UK, where ABA has never before been popular, there are some signs that demand for the therapy has increased, but the reluctance of the NAS to endorse ABA has limited any significant improvement in its status. ABA was developed, and remains popular, in North America, however, where there is more scope for competing theories to co-exist, compared with in the UK; this is in part owing to differences in medical funding. Though organizations such as ASAN and activists such as Michelle Dawson have had public successes in their fight against the pro-cure viewpoint, ABA continues to be widely accepted across the country. The debates about ABA have continued to rage, therefore, in arenas from courtrooms to schools, charity publications to online fora, offering a prism through which to see how definitions of pathology, therapy and abuse have become increasingly contested in relation to autism.

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Notes

1. The title quotation is from Autism Mythbusters (n.d.), *The Truth About ABA* (see References).
2. The *Diagnostic and Statistical Manual of Mental Disorders* (DSM), though often seen to represent a psychiatric consensus, is not universally used for autism diagnoses, even when considering only anglophone countries.
3. Clinical psychologists at the Institute of Psychiatry did also utilize behavioural techniques for the treatment of autism in the late 1980s, but the focus was instead on the use of language-based operant techniques (Howlin and Rutter, 1989).
4. As discussed in Dixon, Vogel and Tarbox (2012); see also for a brief, broad overview of the academic origins of functional analysis, and as well as the work of Fuller, Ferster, and Ayllon and Michael.
5. The ‘refrigerator mother’ was first discussed in Bettelheim (1967), and the idea that mothers caused schizophrenia, at that time seen as closely related to autism, was first discussed in Fromm-Reichman (1948). For further discussion see Staub (2011: 43).
6. The following paragraphs draw heavily on the work on Silberman (2015: 280–8), who recounts the work of Lovaas in detail as part of his broader history of autism. Departing from Silberman, this article takes ABA itself as the central object of exploration, as well as self-advocate responses to the therapy.
7. The ‘Feminine Boy Project’, led by George Rekers, has been discussed in detail by Szalavaitz (2011), Dawson (n.d.) and, in relation to Lovaas, Silberman (2015: 319–23).

8. Professor William Yule, a collaborator on the Institute of Psychiatry's autism research project in the 1980s, has retrospectively argued that ABA had little uptake in the United Kingdom because researchers were skeptical about the evidence for its efficacy (Yule, 2015).
9. The histories of these organizations are discussed in Silverman (2012) and Dolnick (1998).
10. Psychology texts that emphasize differences rather than deficits include, for example, Gardner (2000).
11. The full range of material relating to the JRC produced by ASAN, including accounts from survivors and letters sent to government departments, accessed 21 October 2015, is accessible at: <http://autisticadvocacy.org/tag/jrc/>
12. For a detailed history of the JRC, consult Neumeier (2012).
13. Full details about this court case, accessed 21 October 2015, are accessible at: http://www.sentex.net/~nexus23/naa_sup.html and http://www.sentex.net/~nexus23/naa_fac.html
14. The comparisons mentioned by Dawson are often made by parent-activists rather than self-advocates. Dawson's argument that such non-medical activism has impacted upon ABA research parallels Epstein's arguments that AIDS activists similarly influenced the design of medical trials (Epstein, 1998).

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